

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0758

Billerica 100 kV
December 19, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0758. The Interconnection Customer proposes a 280.9 MW solar photovoltaic facility interconnecting at the Billerica 100 kV substation in the Ohio area. The request was received on March 12, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 260 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,024,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Tejon - Lincoln St West 100 KV	Billerica - Telephone 100 KV L	199.3	112	100.0
Berrywood Substation - Billerica 100 KV Line	Billerica - Telephone 100 KV line	198.1	134	100.0
Lincoln St West - Berrywood 100 KV	Telephone - Billerica 100 KV Line	195.8	128	100.0
Billrica - Telephone 100 KV	Berrywood - Billerica 100 KV L	190.8	139	100.0
Telephone - Shiras 100 KV line	Hyatt - Telephone 100 KV	134.4	102	61.6
Maiposa - Acec Brooks 100 KV	Norwich - Barrett Substation 500 KV	133.1	147	7.7
Neligh - Tejon 100 KV L	Billerica Substation - Telephone 100 KV	131.4	109	49.0
Mariposa Substation - Mosher 100 KV L	Barrett Substation - Norwich 500 KV	127.0	140	7.7
Bonaire - Shiras 100 KV	Telephone - Hyatt 100 KV Line	102.2	119	61.6
Downs Substation - Wilmington Station C Tap Substation 100 KV L	Norwich - Barrett Substation 500 KV L	100.3	137	5.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tejon - Lincoln St West 100 KV	Rebuild line with higher-capacity conductor	\$4,453,000	100% (\$4,453,000)
Berrywood Substation - Billerica 100 KV Line	Rebuild line with higher-capacity conductor	\$1,144,000	100% (\$1,144,000)
Lincoln St W - Berrywood 100 KV	Rebuild line with higher-capacity conductor	\$4,796,000	100% (\$4,796,000)
Billerica - Telephone 100 KV	Rebuild line with higher-capacity conductor	\$3,512,000	100% (\$3,512,000)
Telephone - Shiras 100 KV	Rebuild line with higher-capacity conductor	\$1,438,000	100% (\$1,438,000)
Mariposa - Acec Broos 100 KV	Rebuild line with higher-capacity conductor	\$4,305,000	100% (\$4,305,000)
Tejon - Neligh 100 KV	Rebuild line with higher-capacity conductor	\$3,489,000	100% (\$3,489,000)
Mariposa Substation - Mosher 100 KV L	Rebuild line with higher-capacity conductor	\$3,067,000	100% (\$3,067,000)
Bonaire - Shiras 100 KV	Reconductor line	\$533,000	100% (\$533,000)
Downs Substation - Wilmington Station C Tap Substation 100 KV L	Reconductor line	\$381,000	75% (\$287,000)
	Total allocated to project		\$27,024,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.